

File Handling in Java

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File handling in Java is the process of creating, reading, writing, updating and deleting files using Java programs. Java provides the "**java.io**" and "**java.nio**" packages for file operations.

File handling allows programs to store data permanently in files instead of keeping them temporarily in the memory.

Why File Handling is important?

File handling is used to:

- Store data permanently.
- Read saved data later.
- Manage text and binary files.
- Create reports.
- Save user information.
- Process files in applications.

Java Packages Used for File Handling

Package	Purpose
java.io	Traditional file handling.
java.nio.file	Modern file handling.
java.util	Scanner class for input.

Types of Files

1. **Text Files:** Store data in readable character format.

Examples: .txt, .csv, .java, .py, .html, .css, .js etc.

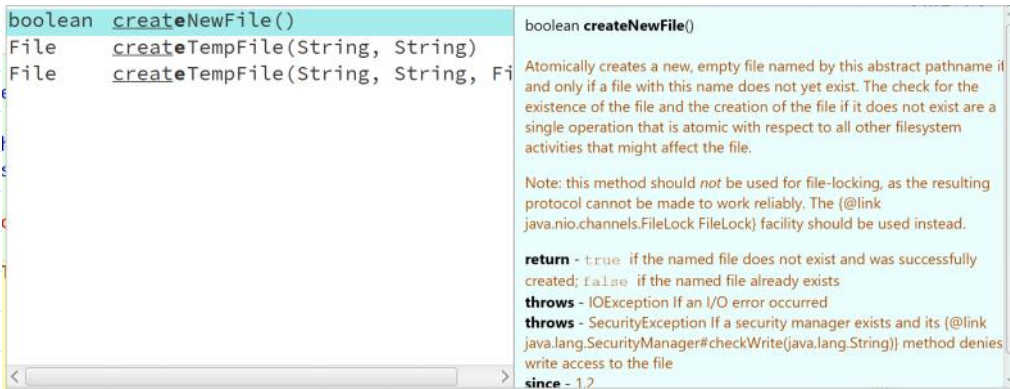
2. **Binary Files:** Store data in byte format. Examples: .dat, .exe, .jpg, .mp4, mp3 etc.

Important Classes Used in File Handling

Class	Purpose
File	Represents a file or directory.
FileWriter	Writes data into files.
FileReader	Reads character files.
BufferedReader	Reads text efficiently.
BufferedWriter	Writes text efficiently.
PrintWriter	Formatted file writing.
Scanner	Read data from files.
FileInputStream	Reads binary data.
FileOutputStream	Writes binary data.

Creating a File in Java

```
boolean createNewFile()  
File createTempFile(String, String)  
Atomicslv creates a new, empty file named by this abstract pathname if
```



```
package FileHandling;
import java.io.*;

/**
 * Write a description of class CreateFileExample here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class CreateFileExample
{
    public static void main(String[] args){
        System.out.print("\fEnter File Name: ");
        String fileName = new java.util.Scanner(System.in).nextLine();
        String path = "./DataFiles/";
        try{
            File file = new File(path+fileName);
            if(file.createNewFile()){
                System.out.print("\nFile created successfully.");
                System.out.print("\nFile name: " + file.getName());
            }
            else{
                System.out.print("\nFile already exists.");
            }
        }catch(IOException e){
            System.err.print("\nError while creating a file.");
            e.printStackTrace();
        }
    }
}
```

Relative path associated with current working project folder.

Output:

☐ Name	Date modified	Type	Size
FirstFile.txt	11-05-2026 20:54	txtfilelegacy	0 KB
two.java	11-05-2026 20:55	JAVA File	0 KB
three.py	11-05-2026 20:56	PyCharmCE2021.2	0 KB

To create directory or folder in our project folder we can use this code:

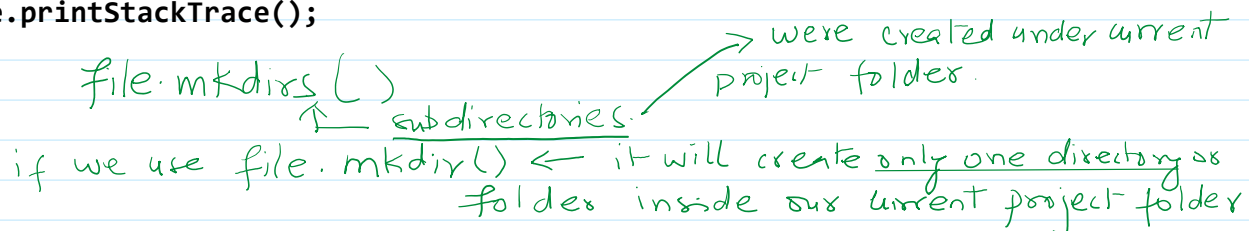
```
package FileHandling;
```

```

import java.io.*;

/**
 * Write a description of class CreateDirectoryExample here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class CreateDirectoryExample
{
    public static void main(String[] args){
        System.out.print("\nEnter directory Name: ");
        String directoryName = new java.util.Scanner(System.in).nextLine();
        String path = "./";
        try{
            File file = new File(path+directoryName);
            if(file.mkdirs()){
                System.out.print("\nDirectory created successfully.");
                System.out.print("\nFile name: " + file.getAbsolutePath());
            }
            else{
                System.out.print("\nFile already exists.");
            }
        }catch(SecurityException e){
            System.err.print("\nError while creating a file.");
            e.printStackTrace();
        }
    }
}

```

`file.mkdirs()` 

 ↑ subdirectories.

if we use `file.mkdir()` ← it will create only one directory or folder inside our current project folder.